Band-1: Lang–Trotter — Single Non-CM Curve Error Probe (collapse 2.23)

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Honesty stamp. Increment / ledger only. **Does not** solve Lang–Trotter. **No** new proven error exponent. **No** Clay stamp. **No** ToE. **GATE-04 LOCKED.**

Lang–Trotter remains OPEN for any single non-CM curve.

Abstract

Pick one non-CM elliptic curve over $\mathbb{Q}$ — Cremona **11a1** — and ledger finite counts $\pi_{E,r}(X) = \#\{p \le X : a_p(E) = r\}$ against the conjectural **order** $\sqrt{X}/\log X$. Box bound capped at $X = 5 \cdot 10^3$ (668 good primes; wall ≈ 0.81 s). Literature upper bounds (Murty; Elkies; Zywina under GRH) and average results (Fouvry–Murty; Baier) are cited; **no individual asymptotic is claimed**.

Curve

Field	Value
Cremona	**11a1**
LMFDB	11.a3
Model	$y^2 + y = x^3 - x^2 - 10x - 20$
Conductor	11
CM?	**No**

Literature ledger (not re-proved on box)

Item	Statement	Status
Conjecture (Lang–Trotter 1976 LNM 504)	$\pi_{E,r}(x) \sim C_{E,r}\sqrt{x}/\log x$	**OPEN** (individual)
Uncond. $r \neq 0$	$\pi_{E,r}(x) \ll_E x(\log\log x)^2/(\log x)^2$	V.K. Murty 1997
Uncond. $r = 0$	$\pi_{E,0}(x) \ll_E x^{3/4}$	Elkies–Kaneko–Murty
GRH	$\ll_E x^{4/5}/(\log x)^{3/5}$; $r = 0$: $x^{3/4}/(\log x)^{1/2}$	Zywina arXiv:1508.07682

Item	Statement	Status
Average	asymptotics over $E(a,b)$ in boxes	Fouvry–Murty; David–Pappalardi; Baier arXiv:math/0609095

Gap this probe does not close: individual asymptotic / improved individual error for a fixed non-CM E.

Method (box)

1. For each prime $p \le X = 5000$, $p \neq 11$: compute $\#E(\mathbb{F}_p)$ by Legendre sum on the completed-square model; set $a_p = p + 1 - \#E$.
2. Spot-check $a_2, a_3, a_5, a_7, a_{13}, a_{17}, a_{19}$ against the known Cremona table (all match).
3. Hasse check $|a_p| \le 2\sqrt{p}$ for all 668 good primes (pass).
4. Ledger $\pi_{E,r}(X)$ and ratios $\pi_{E,r}(X)/(\sqrt{X}/\log X)$ (empirical proxies for $C_{E,r}$; **not** proofs).
5. Nested windows at $500, 1000, 2000, 5000$.

Results

Order comparison at $X = 5000$

$\sqrt{X}/\log X \approx 8.302$. Good primes: **668**.

r	$\pi_{E,r}(X)$	ratio vs $\sqrt{X}/\log X$
-2	11	≈ 1.325
-1	6	≈ 0.723
0	11	≈ 1.325
1	1	≈ 0.120
2	10	≈ 1.205

Nested growth ($r = 0$)

X	$\pi_{E,0}$	$\sqrt{X}/\log X$	ratio
500	3	≈ 3.60	≈ 0.83
1000	5	≈ 4.58	≈ 1.09
2000	7	≈ 5.88	≈ 1.19
5000	11	≈ 8.30	≈ 1.32

Counts track the **order** of $\sqrt{x}/\log x$ at this tiny scale; that is consistent with (not a proof of) Lang–Trotter. Sparse $r = 1$ at this X is expected when $C_{E,1}$ is small / fluctuations dominate.

Spot checks

Known table $a_2 = -2$, $a_3 = -1$, $a_5 = 1$, $a_7 = -2$, $a_{13} = 4$, $a_{17} = -2$, $a_{19} = 0$ — **all matched**.

Why Band-1 vs conjecture

- **Band-1 (this probe):** cite known bounds/averages; certify a finite a_p histogram for one non-CM curve; order ledger.
- **Conjecture (OPEN):** $\pi_{E,r}(x) \sim C_{E,r}\sqrt{x}/\log x$ for fixed non-CM E — **not proved here**.
- **Improved error (OPEN):** no new individual upper-bound exponent tonight.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/lang_trotter/`
- Run: `python3 lang_trotter_probe.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`

Conclusion

Curve: Cremona **11a1** (non-CM). **Box:** a_p histogram for primes $\leq 5{\cdot}10^3$ (capped). **Lang–Trotter: OPEN**. No improved error theorem. No Clay. No ToE. **GATE-04 LOCKED**.